1. CONFIGURACION Y CONECCION DE TRES ROUTERS, TRES SWITCHES Y DOS PC EN CADA SHUTCHES PARA UNA EMPRESA:

1. EN EL RUTER DE CLARO HACEMOS LA CONEXION EN GIGABITE 0/1 AL SWITCH
2. EN EL RUTER DE VIVA HACEMOS LA CONEXION EN GIGABITE 0/1 AL SWITCH
3. EN EL RUTER DE ALTICE HACEMOS LA CONEXION EN GIGABITE 0/1 AL SWITCH

2. AHORA HACEMOS LA CONECCION DE LOS 3 ROUTERS:

1. EL DE CLARO LO CONECTAMOS EN EL DE GIGABITE 1 AL DE GIGABITE 1 DE VIVA Y EL DE VIVA LO CONECTAMOS AL GIGABITE 2 AL UNO DE ALTICE.

3.CONECCIONES DE LOS SHITCH CLARO, VIVA, ALTICE:

1.CLARO

AHORA CONECTAMOS EL SWITCH DE CLARO EN FAST ETHERNET 1 AL FAST ETHERNET 0 EN LA PC 1

AHORA CONECTAMOS EL SWITCH DE CLARO EN FAST ETHERNET 2 AL FAST ETHERNET 0 EN LA PC 2

2.VIVA

AHORA CONECTAMOS EL SWITCH DE VIVA EN FAST ETHERNET 1 AL FAST ETHERNET 0 EN LA PC 3

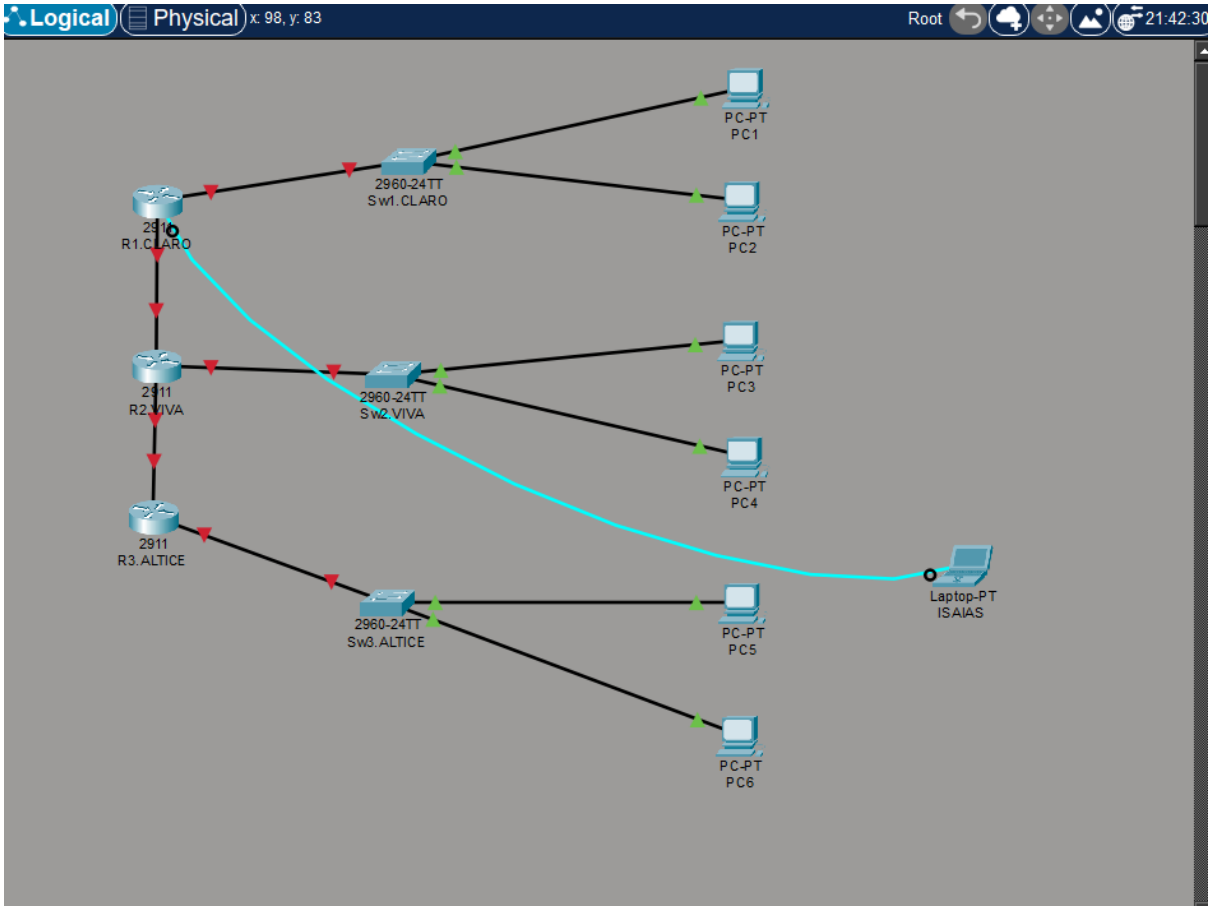
AHORA CONECTAMOS EL SWITCH DE VIVA EN FAST ETHERNET 2 AL FAST ETHERNET 0 EN LA PC 4

3.ALTICE

AHORA CONECTAMOS EL SWITCH DE ALTICE EN FAST ETHERNET 1 AL FAST ETHERNET 0 EN LA PC 5

AHORA CONECTAMOS EL SWITCH DE ALTICE EN FAST ETHERNET 2 AL FAST ETHERNET 0 EN LA PC 6

YA TENEMOS TODAS LAS CONECCIONES :



4. AHORA VAMOS A CONFIGURAR CONFIGURAR TODOD LOS EQUIPOS

1. R1.CLARO

en

conf t

no ip domain-lookup

host isaias

ip domain-name hackdark.com

enable secret h@ckd4rk159*

username isaias privilege 15 secret f1n@l666*

crypto key generate rsa

1024

ip ssh version 2

ip ssh authentication-retries 3

ip ssh time-out 60

line con 0

pass w0rr13-

logging synchronous

login

line vty 0 15

logging synchronous

login local

transport input ssh

no transport input telnet

end

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no ip domain-lookup
Router(config)#host isaias
isaias(config)#ip domain-name hackdark.com
isaias(config)#enable secret h@ckd4rk159*
isaias(config)#username isaias privilege 15 secret f1n@l666*
isaias(config)#crypto key generate rsa
The name for the keys will be: isaias.hackdark.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

isaias(config)#ip ssh version 2
*Mar 1 1:36:10.613: %SSH-5-ENABLED: SSH 1.99 has been enabled
isaias(config)#ip ssh authentication-retries 3
isaias(config)#ip ssh time-out 60
isaias(config)#line con 0
isaias(config-line)#pass w0rr13-
isaias(config-line)#logging synchronous
isaias(config-line)#login
isaias(config-line)#line vty 0 15
isaias(config-line)#logging synchronous
isaias(config-line)#login local
isaias(config-line)#transport input ssh
isaias(config-line)#no transport input telnet
isaias(config-line)#end
isaias#
%SYS-5-CONFIG_I: Configured from console by console

isaias#
isaias#
isaias#
```

2.R2.VIVA

```
en
conf t
no ip domain-lookup
host isaias
ip domain-name hackdark.com
enable secret h@ckd4rk159*
username isaias privilege 15 secret f1n@l666*
crypto key generate rsa
1024
ip ssh version 2
ip ssh authentication-retries 3

ip ssh time-out 60
line con 0
pass w0rr13-
logging synchronous
login
line vty 0 15
logging synchronous
login local
transport input ssh
no transport input telnet
end
```

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no ip domain-lookup
Router(config)#host isaias
isaias(config)#ip domain-name hackdark.com
isaias(config)#enable secret h@ckd4rk159*
isaias(config)#username isaias privilege 15 secret f1n@l666*
isaias(config)#crypto key generate rsa
The name for the keys will be: isaias.hackdark.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

isaias(config)#ip ssh version 2
*Mar 1 1:36:10.613: %SSH-5-ENABLED: SSH 1.99 has been enabled
isaias(config)#ip ssh authentication-retries 3
isaias(config)#ip ssh time-out 60
isaias(config)#line con 0
isaias(config-line)#pass w0rr13-
isaias(config-line)#logging synchronous
isaias(config-line)#login
isaias(config-line)#line vty 0 15
isaias(config-line)#logging synchronous
isaias(config-line)#login local
isaias(config-line)#transport input ssh
isaias(config-line)#no transport input telnet
isaias(config-line)#end
isaias#
%SYS-5-CONFIG_I: Configured from console by console

isaias#
isaias#
isaias#
```

3. R3.ALTICE

en

conf t

no ip domain-lookup

host isaias

ip domain-name hackdark.com

enable secret h@ckd4rk159*

username isaias privilege 15 secret f1n@l666*

crypto key generate rsa

1024

ip ssh version 2

ip ssh authentication-retries 3

ip ssh time-out 60

line con 0

pass w0rr13-

logging synchronous

login

line vty 0 15

logging synchronous

login local

transport input ssh

no transport input telnet

end

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no ip domain-lookup
Router(config)#host isaias
isaias(config)#ip domain-name hackdark.com
isaias(config)#enable secret h@ckd4rk159*
isaias(config)#username isaias privilege 15 secret f1n@1666*
isaias(config)#crypto key generate rsa
The name for the keys will be: isaias.hackdark.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

isaias(config)#ip ssh version 2
*Mar 1 1:37:48.446: %SSH-5-ENABLED: SSH 1.99 has been enabled
isaias(config)#ip ssh authentication-retries 3
isaias(config)#ip ssh time-out 60
isaias(config)#line con 0
isaias(config-line)#pass w0rr13-
isaias(config-line)#logging synchronous
isaias(config-line)#login
isaias(config-line)#line vty 0 15
isaias(config-line)#logging synchronous
isaias(config-line)#login local
isaias(config-line)#transport input ssh
isaias(config-line)#no transport input telnet
isaias(config-line)#end
isaias#
%SYS-5-CONFIG_I: Configured from console by console

isaias#
isaias#
isaias#
```

4. Sw1.CLARO

```
en
conf t
no ip domain-lookup
host isaias
ip domain-name hackdark.com
enable secret h@ckd4rk159*
username isaias privilege 15 secret f1n@l666*
crypto key generate rsa
1024
ip ssh version 2
ip ssh authentication-retries 3
ip ssh time-out 60
line con 0
pass w0rr13-
logging synchronous
login
line vty 0 15
logging synchronous
login local
transport input ssh
no transport input telnet
end
```

```
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#no ip domain-lookup
Switch(config)#host isaias
isaias(config)#ip domain-name hackdark.com
isaias(config)#enable secret h@ckd4rk159*
isaias(config)#username isaias privilege 15 secret f1n@l666*
isaias(config)#crypto key generate rsa
The name for the keys will be: isaias.hackdark.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

isaias(config)#ip ssh version 2
*Mar 1 1:41:3.213: %SSH-5-ENABLED: SSH 1.99 has been enabled
isaias(config)#ip ssh authentication-retries 3
isaias(config)#ip ssh time-out 60
isaias(config)#line con 0
isaias(config-line)#pass w0rr13-
isaias(config-line)#logging synchronous
isaias(config-line)#login
isaias(config-line)#line vty 0 15
isaias(config-line)#logging synchronous
isaias(config-line)#login local
isaias(config-line)#transport input ssh
isaias(config-line)#no transport input telnet
isaias(config-line)#end
isaias#
%SYS-5-CONFIG_I: Configured from console by console

isaias#
isaias#
isaias#
```

5. Sw2.CLARO

en

conf t

no ip domain-lookup

host isaias

ip domain-name hackdark.com

enable secret h@ckd4rk159*

username isaias privilege 15 secret f1n@l666*

crypto key generate rsa

1024

```
ip ssh version 2
ip ssh authentication-retries 3
ip ssh time-out 60
line con 0
pass w0rr13-
logging synchronous
login
line vty 0 15
logging synchronous
login local
transport input ssh
no transport input telnet
end
```

```
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#no ip domain-lookup
Switch(config)#host isaias
isaias(config)#ip domain-name hackdark.com
isaias(config)#enable secret h@ckd4rk159*
isaias(config)#username isaias privilege 15 secret f1n@l666*
isaias(config)#crypto key generate rsa
The name for the keys will be: isaias.hackdark.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

isaias(config)#ip ssh version 2
*Mar 1 1:41:3.213: %SSH-5-ENABLED: SSH 1.99 has been enabled
isaias(config)#ip ssh authentication-retries 3
isaias(config)#ip ssh time-out 60
isaias(config)#line con 0
isaias(config-line)#pass w0rr13-
isaias(config-line)#logging synchronous
isaias(config-line)#login
isaias(config-line)#line vty 0 15
isaias(config-line)#logging synchronous
isaias(config-line)#login local
isaias(config-line)#transport input ssh
isaias(config-line)#no transport input telnet
isaias(config-line)#end
isaias#
%SYS-5-CONFIG_I: Configured from console by console

isaias#
isaias#
isaias#
```

6. Sw3.CLARO

en

conf t

no ip domain-lookup

host isaias

ip domain-name hackdark.com

enable secret h@ckd4rk159*

username isaias privilege 15 secret f1n@l666*

crypto key generate rsa

1024

ip ssh version 2

```
ip ssh authentication-retries 3
ip ssh time-out 60
line con 0
pass w0rr13-
logging synchronous
login
line vty 0 15
logging synchronous
login local
transport input ssh
no transport input telnet
end
```

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#no ip domain-lookup
Switch(config)#host isaias
isaias(config)#ip domain-name hackdark.com
isaias(config)#enable secret h@ckd4rk159*
isaias(config)#username isaias privilege 15 secret f1n@1666*
isaias(config)#crypto key generate rsa
The name for the keys will be: isaias.hackdark.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

isaias(config)#ip ssh version 2
*Mar 1 1:42:29.81: %SSH-5-ENABLED: SSH 1.99 has been enabled
isaias(config)#ip ssh authentication-retries 3
isaias(config)#ip ssh time-out 60
isaias(config)#line con 0
isaias(config-line)#pass w0rr13-
isaias(config-line)#logging synchronous
isaias(config-line)#login
isaias(config-line)#line vty 0 15
isaias(config-line)#logging synchronous
isaias(config-line)#login local
isaias(config-line)#transport input ssh
isaias(config-line)#no transport input telnet
isaias(config-line)#end
isaias#
%SYS-5-CONFIG_I: Configured from console by console

isaias#
isaias#
isaias#
isaias#
```

Configuracion basica de Interfas en Routers Cisco:

FORMMA LARGA DE COMANDOS:

enable : modo privilegiado
configure terminal : entramos al modo de configuracion gloval
interface ggabiteEthernet0/1 : entramos al modo de la interfaz que deseamos configurar
description : le agregamos una descripcion a la interfaz (util para describir la configuracion que se implementa en dicha interfaz)
ip address [direccion-ip-asignada] [mascara-subred]: le agregamos la direccion ip asignada (segun el plan de direccionamiento ip) con la mascara de subred correspondiente
(tambien siguiendo el plan de direcionamiento ip)
no shutdown (encendemos manualmente la interfaz de red para establecer comunicacion)

forma corta:

en :entramos al modo privilegiado
conf t :entramos al modo de configuracion global
int [id-interfaz] :entramos al modo de configuracion de la interfaz que deseamos configurar
desc : le agregamos una descripcion a la interfaz (util para describir la configuracion que se implementa en dicha interfaz)
ip add [direccion-ip-asignada] [mascara-subred] :le agregamos la direccion ip asignada (segun el plan de direccionamiento IP) con la
mascara de subred correspondiente (tambien siguiendo el plan de direccionamiento IP)
no shut (encendemos manualmente la interfaz de red para establecer comunicacion)

ejemplo:

```
en
conf t
int g0/1
desc Red LAN Ventas
ip add 192.168.0.1 255.255.255.0
no shut
```

Ojo: Recordar que al finalizar la configuracion, debemos ir al modo privilegiado y guardar los cambios.

Plan de Direccionamiento IPv4 para las Redes en el LAB3

Red LAN Ventas:
IP-Red: 192.168.38.0
Broadcast: 192.168.38.63
Mascara: 255.255.255.192
prefijo: /26
Rango: 192.168.38.1 - 192.168.38.62
IP-Router: 192.168.38.1

Red LAN Compras:
IP-Red: 192.168.38.192
Broadcast: 192.168.38.223
Mascara: 255.255.255.224
prefijo: /27
Rango: 192.168.38.193 - 192.168.38.222
IP-Router: 192.168.38.193

Red LAN Almacen:
IP-Red: 192.168.38.224
Broadcast: 192.168.38.239
Mascara: 255.255.255.240
prefijo: /28
Rango: 192.168.38.225 - 192.168.38.238
IP-Router: 192.168.38.225
Red WAN Ventas <- > Compras:
IP-Red: 192.168.38.240
Broadcast: 192.168.38.243

Mascara: 255.255.255.252

1.Router Ventas:

```
h@ckd4rk159*
en
f1n@l666*
conf t
int g0/1
desc Red LAN Ventas
ip add 192.168.38.1 255.255.255.192
no shut
int g0/0
desc Red WAN: Ventas -> Compras
ip add 192.168.38.241 255.255.255.252
no shut
end
copy run start
show run
```

2.router contabilidad

```
h@ckd4rk159*
en
f1n@l666*
conf t
int g0/1
desc Red LAN Ventas
ip add 192.168.38.193 255.255.255.224
no shut
int g0/01
desc Red WAN: Ventas -> Compras
ip add 192.168.38.193 192.168.38.222
no shut
end
copy run start
show run
```

3. R3.SOPORTE-TI

```
h@ckd4rk159*
en
f1n@l666*
conf t
int g0/1
desc Red LAN Ventas
ip add 192.168.38.225 255.255.255.240
no shut
int g0/2
desc Red WAN: Ventas -> Compras
ip add 192.168.38.225 192.168.38.238
no shut
end
```

```
copy run start
show run
```

4. Sw1.CONTABILIDAD

```
h@ckd4rk159*
en
f1n@l666*
conf t
int g0/0
desc Red LAN Ventas
ip add 192.168.38.241 255.255.255.252
no shut
int g0/1
desc Red WAN: Ventas -> Compras
ip add 192.168.38.243 192.168.38.242
no shut
end
copy run start
show run
```

5. Sw2.VENTAS

```
h@ckd4rk159*
en
f1n@l666*
conf t
int g0/1
desc Red LAN Ventas
ip add 192.168.38.244 255.255.255.252
no shut
int g0/0
desc Red WAN: Ventas -> Compras
ip add 192.168.38.245 192.168.38.246
no shut
end
copy run start
show run
```

Documentacion LAB#3 - Isaias Recio.

#1. documentacion de conexiones:

redes LAN:

1. ventas:
   - SW: g0/1
   - R1: g0/1

2. compras:
   - SW: g0/1
   - R1: g0/1

3. almacen:
      SW: g0/1
      R1: g0/1

redes WAN:

1. venta - compra:
      ventas: g0/0 172.16.0.1
      compras: g0/3 172.16.0.2

2. compra - almacen:
      compras: g0/2
      almacen: g0/2

1. Router Ventas (R1)

```
w0rr13-
h@ckd4rk159*
f1n@l666*
enable
configure terminal
hostname R-VENTAS

interface g0/1
 description LAN_VENTAS
 ip address 192.168.38.1 255.255.255.192
 no shutdown

interface g0/0
 description WAN_HACIA_CONTABILIDAD
 ip address 192.168.38.241 255.255.255.252
 no shutdown

# Rutas Estáticas: Para llegar a Contabilidad y Soporte
ip route 192.168.38.192 255.255.255.224 192.168.38.242
ip route 192.168.38.224 255.255.255.240 192.168.38.242
do write
```

2. Router Contabilidad (R2) - El Central

```
enable
configure terminal
hostname R-CONTABILIDAD

interface g0/1
 description LAN_CONTABILIDAD
 ip address 192.168.38.193 255.255.255.224
 no shutdown

interface g0/0
 description WAN_HACIA_VENTAS
 ip address 192.168.38.242 255.255.255.252
```

```
 no shutdown

interface g0/2
 description WAN_HACIA_SOPORTE
 ip address 192.168.38.245 255.255.255.252
 no shutdown

# Rutas Estáticas: Para llegar a los extremos
ip route 192.168.38.0 255.255.255.192 192.168.38.241
ip route 192.168.38.224 255.255.255.240 192.168.38.246
do write
```

3. Router Soporte-TI (R3)
El último extremo de la red.

```
Bash
enable
configure terminal
hostname R-SOPORTE

interface g0/1
 description LAN_SOPORTE
 ip address 192.168.38.225 255.255.255.240
 no shutdown

interface g0/0
 description WAN_HACIA_CONTABILIDAD
 ip address 192.168.38.246 255.255.255.252
 no shutdown

# Rutas Estáticas: Para volver a Ventas y Contabilidad
ip route 192.168.38.0 255.255.255.192 192.168.38.245
ip route 192.168.38.192 255.255.255.224 192.168.38.245
do write
```

1. Switch Ventas (SW-VENTAS)
Este switch está en la red 192.168.38.0 /26. Su "jefe" es el Router Ventas (.1).

```
Bash
enable
configure terminal
hostname SW-VENTAS

interface vlan 1
 description IP_ADMINISTRACION
 ip address 192.168.38.2 255.255.255.192
 no shutdown
 exit

ip default-gateway 192.168.38.1

do write
```

2. Switch Contabilidad (SW-CONTABILIDAD)
Este switch está en la red 192.168.38.192 /27. Su "jefe" es el Router Contabilidad (.193).

Bash
```
enable
configure terminal
hostname SW-CONTABILIDAD

interface vlan 1
 description IP_ADMINISTRACION
 ip address 192.168.38.194 255.255.255.224
 no shutdown
 exit

ip default-gateway 192.168.38.193

do write
```

3. Switch Soporte-TI (SW-SOPORTE)
Este switch está en la red 192.168.38.224 /28. Su "jefe" es el Router Soporte (.225).

Bash
```
enable
configure terminal
hostname SW-SOPORTE

interface vlan 1
 description IP_ADMINISTRACION
 ip address 192.168.38.226 255.255.255.240
 no shutdown
 exit

ip default-gateway 192.168.38.225

do write
```